COMPUTER VISION AND IMAGE ANALYSIS

Computer vision is a field of artificial intelligence that enables computers to interpret and understand visual information from the world. It involves developing algorithms and models to process, analyze, and make sense of images and videos.

COMPUTER VISION

AREAS

Image Processing: Enhancing and manipulating images to improve their quality or extract useful information. This includes operations like filtering, resizing, and transforming images.

Object Detection and Recognition: Identifying and locating objects within images or videos. This can be used for tasks like face detection, pedestrian detection, and recognizing specific objects.

Image Segmentation: Dividing an image into meaningful regions or segments, often used to isolate different parts of an image, such as separating foreground from background.

Feature Extraction: Identifying and extracting important features from images, such as edges, corners, and textures, which can be used for further analysis or recognition.

Image Classification: Assigning a label or category to an entire image based on its content, such as determining whether an image contains a cat or a dog.

Pattern Recognition: Detecting patterns and regularities in images, which can be used for tasks like fingerprint recognition or identifying defects in manufacturing.

Motion Analysis: Analyzing movement in videos, which can be used for applications like tracking objects, gesture recognition, and activity monitoring.

VIDEOS

https://www.youtube.com/watch?v=puB-4LuRNys&t=182s

https://www.youtube.com/watch?v=oGvHtpJMO3M

HISTORY

https://medium.com/@ambika199820/what-is-computer-vision-history-applications-challenges-13f5759b48a5

https://letsdatascience.com/learn/history/history-of-computer-vision/

https://uploads-ssl.webflow.com/6436d74a13b2f4f107d23a7d/6464eb11ea28204d6a1eb9e8_1676571719313.jpeg

IMAGE ANALYSIS

Image analysis involves processing and interpreting images to extract meaningful information. It encompasses a wide range of techniques and applications,

IMAGE ANALYSIS

1. **Image Processing**: Enhancing and transforming images using techniques like filtering, edge detection, and noise reduction.
2. **Feature Extraction**: Identifying and isolating specific features within an image, such as edges, corners, or textures.
3. **Object Detection and Recognition**: Identifying and classifying objects within an image, such as detecting faces or recognizing handwritten digits.
4. **Segmentation**: Dividing an image into meaningful regions or segments, often used in medical imaging to isolate different tissues or organs.
5. **Image Classification**: Assigning a label or category to an entire image, such as determining whether an image contains a cat or a dog.
6. **Pattern Recognition**: Identifying patterns and regularities in images, which can be used for tasks like fingerprint recognition or identifying defects in manufacturing.

PACKAGES

1. **OpenCV**: A comprehensive library for computer vision tasks. It supports a wide range of image processing operations, from basic transformations to complex object detection and recognition.
2. **Pillow**: An easy-to-use library for basic image processing tasks like opening, manipulating, and saving different image file formats.
3. **scikit-image**: A collection of algorithms for image processing, built on top of SciPy. It's great for tasks like filtering, segmentation, and feature extraction.
4. **TensorFlow and PyTorch**: These deep learning frameworks are often used for more advanced image analysis tasks, such as image classification, object detection, and image generation.

IMAGE CLASSIFICATION

Image classification is the process of categorizing and labeling groups of pixels or vectors within an image based on specific rules. It's crucial for tasks such as facial recognition, medical imaging, and autonomous driving.

https://www.superannotate.com/blog/image-classification-basics

HOW DOES IMAGE CLASSIFICATION WORK?

1. Data Collection

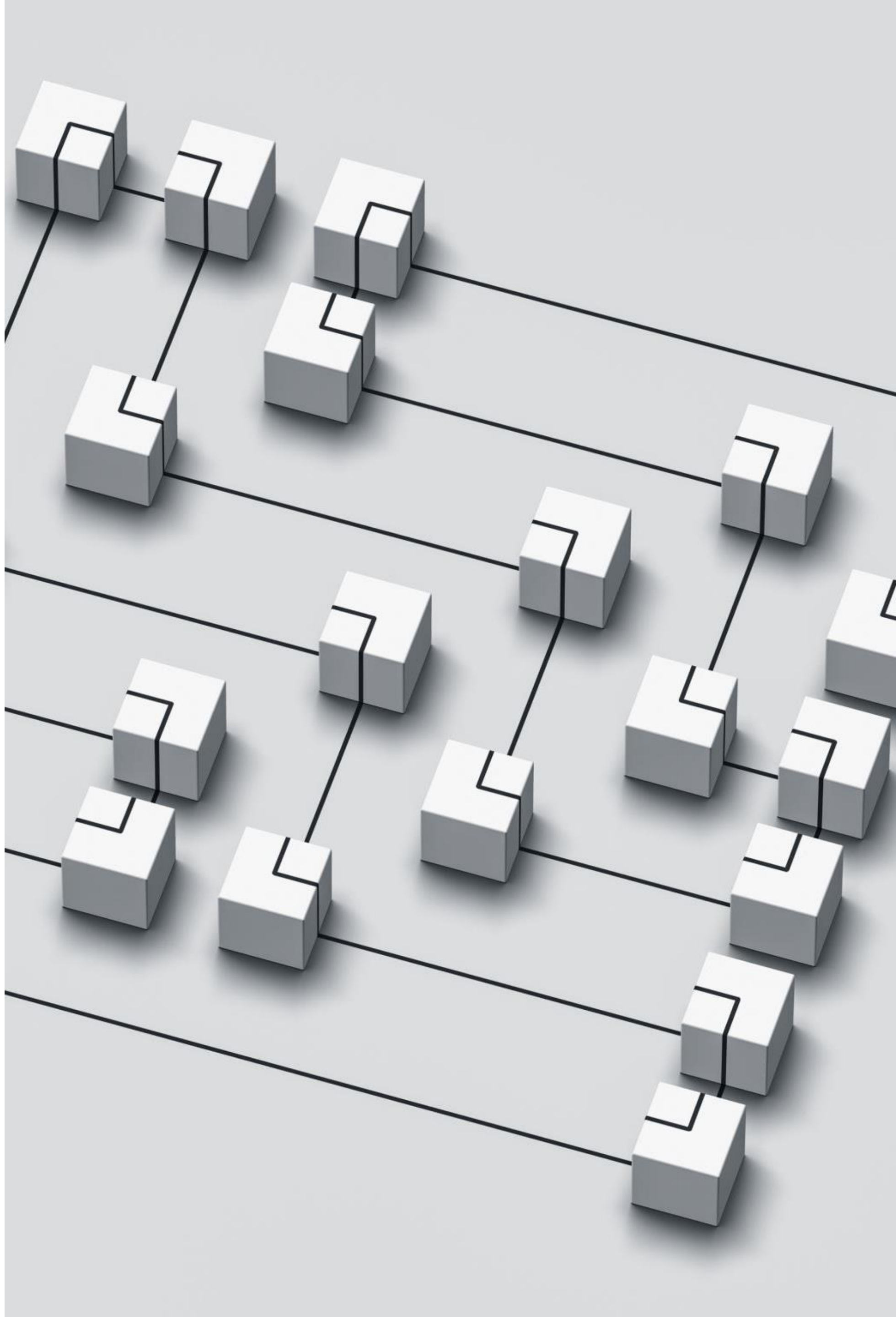
- **Gathering Images:** Collect a large dataset of images, each labeled with the correct category (e.g., cats, dogs, cars).

2. Preprocessing

- **Image Resizing:** Resize images to a uniform size to ensure consistency.
- **Normalization:** Adjust pixel values to a standard range (e.g., 0 to 1) to improve model performance.

3. Feature Extraction

- **Manual Feature Extraction:** Traditional methods involve manually defining features like edges, textures, and shapes.
- **Automatic Feature Extraction:** Modern methods use deep learning models, such as Convolutional Neural Networks (CNNs), to automatically learn features from raw pixel data.

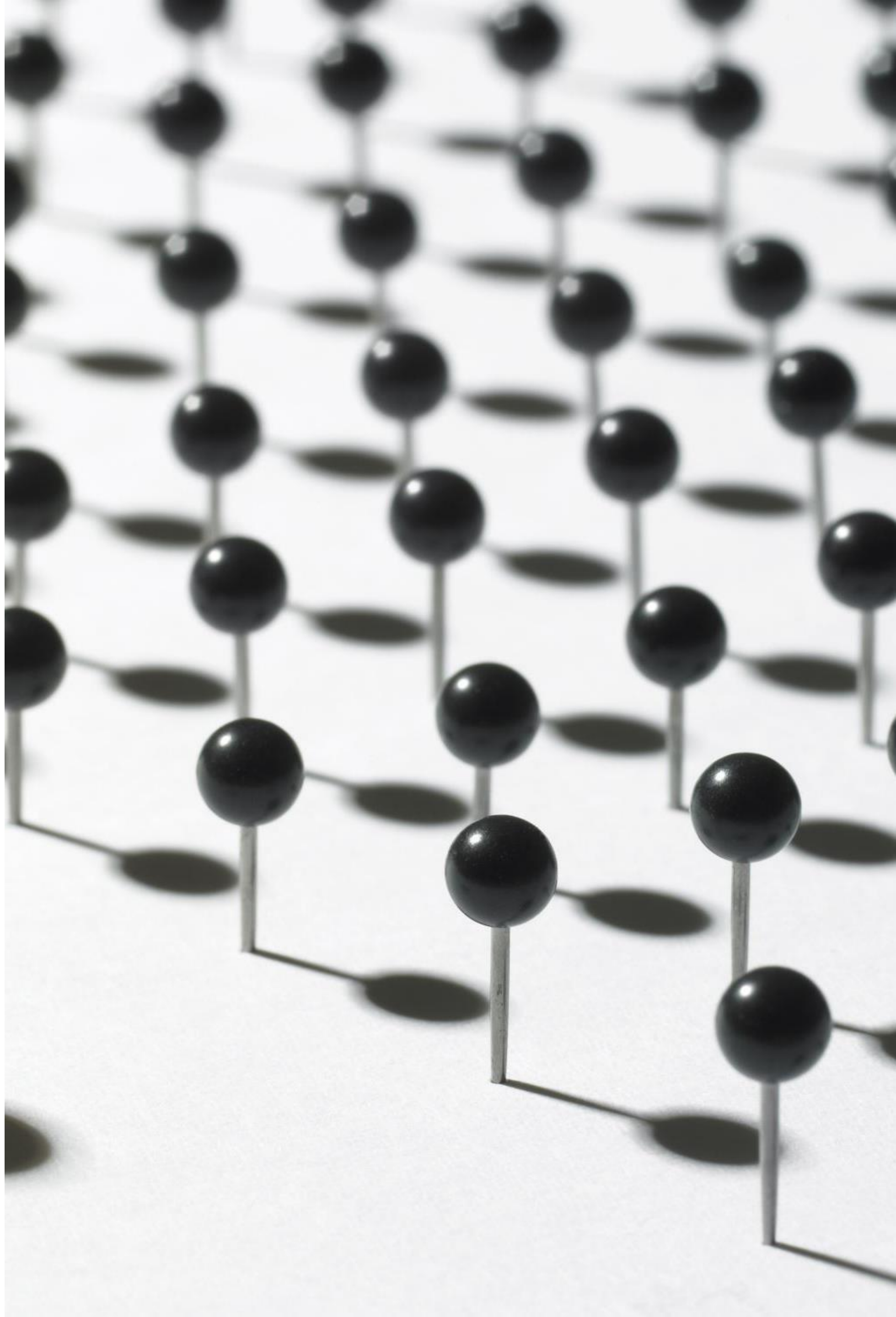


4. Model Training

- **Choosing a Model:** Select a suitable model architecture (e.g., CNNs).
- **Training the Model:** Feed the labeled images into the model, allowing it to learn patterns and features that distinguish different categories. This involves adjusting model parameters to minimize classification errors.

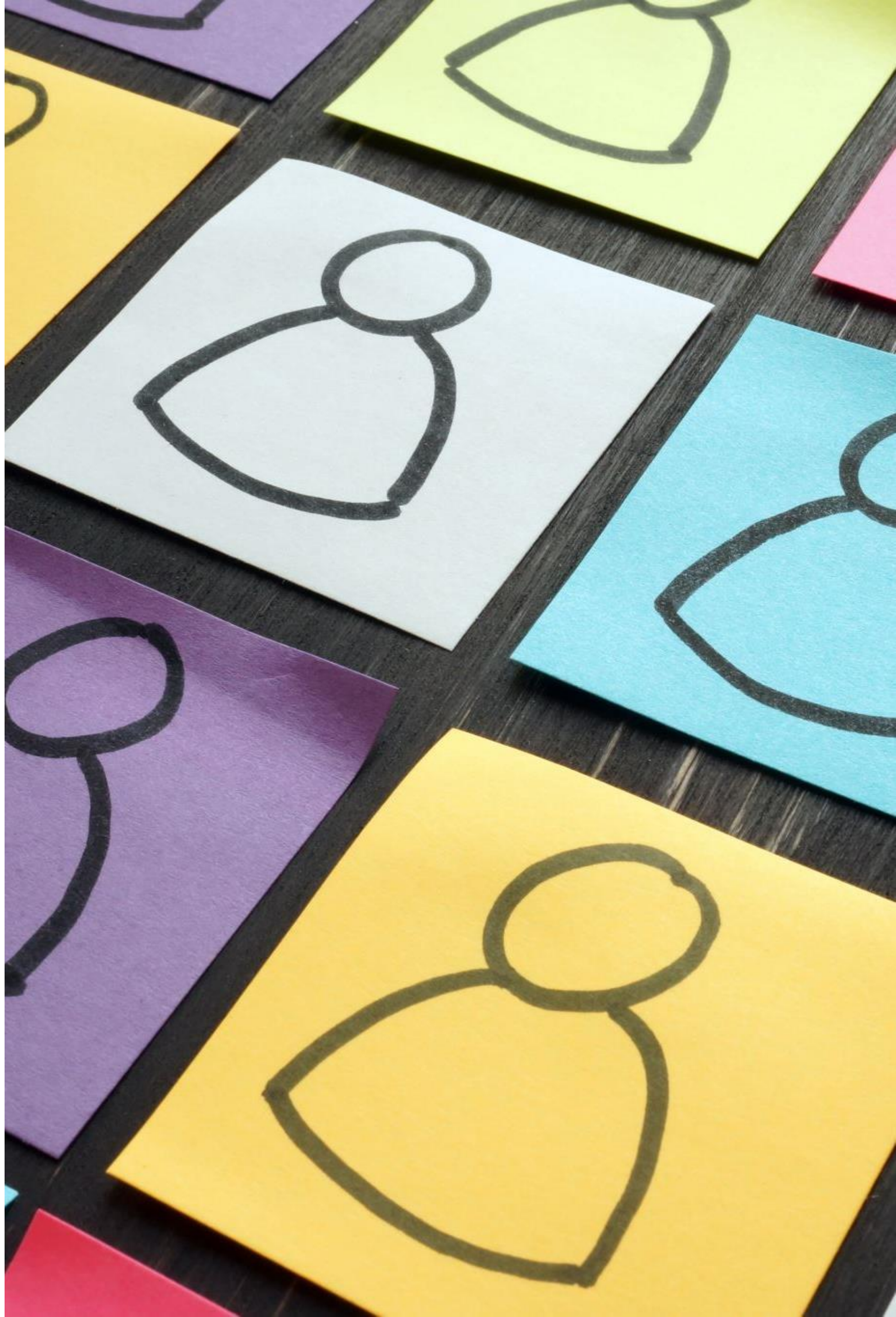
5. Evaluation

- **Testing:** Evaluate the model's performance on a separate set of images (test set) that were not used during training.
- **Metrics:** Use metrics like accuracy, precision, recall, and F1-score to measure how well the model performs.



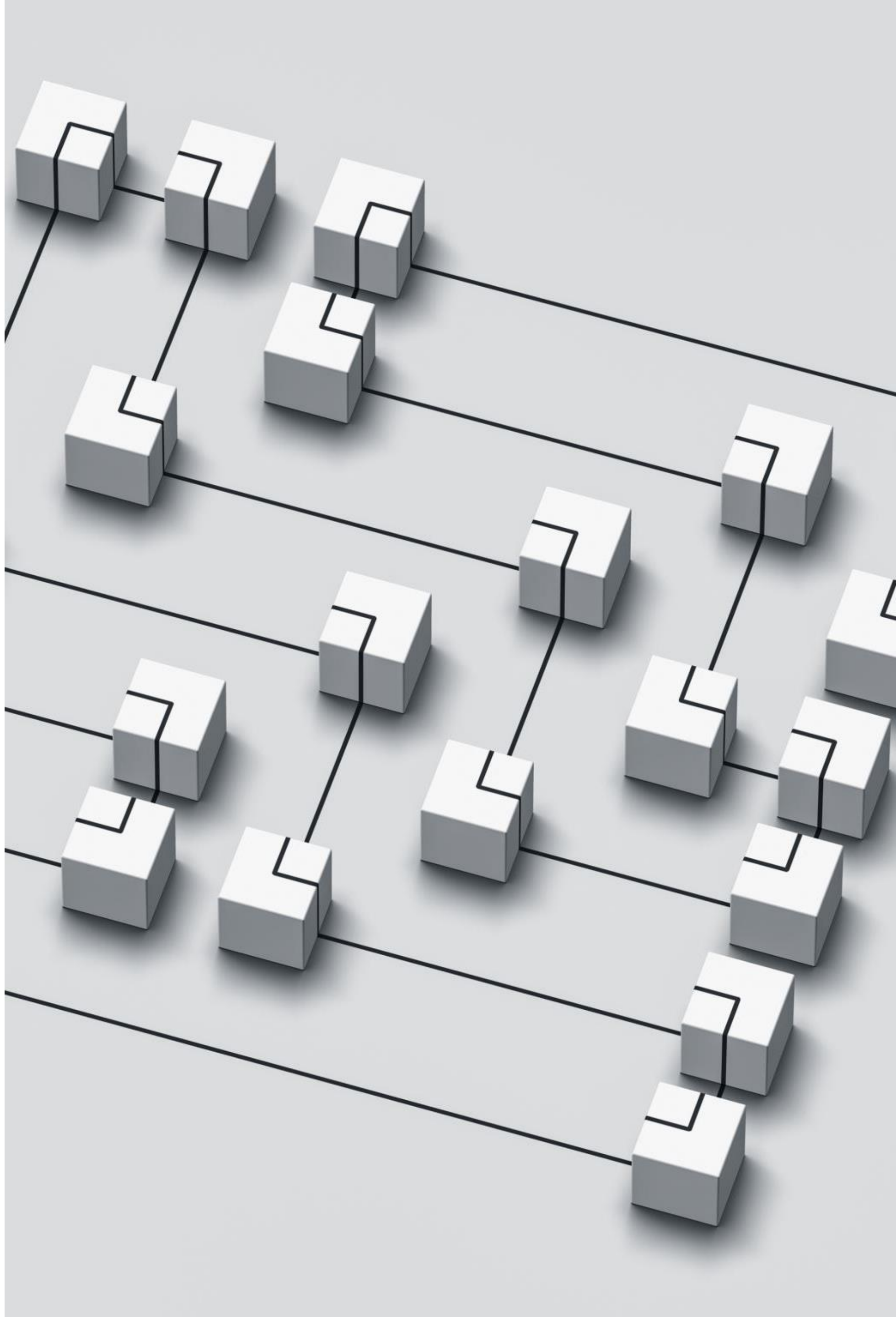
6. Prediction

- **Classifying New Images:** Once trained, the model can classify new, unseen images by predicting the category label based on learned features.



7. Deployment

- **Using the Model:** Deploy the trained model in real-world applications, such as image search engines, autonomous vehicles, or medical diagnosis systems.



APPLICATIONS

1. Healthcare

- **Medical Imaging**: Used to diagnose diseases by classifying medical images such as X-rays, MRIs, and CT scans.
- **Pathology**: Helps in identifying and classifying cells and tissues in pathology slides.

2. Security

- **Facial Recognition**: Used in surveillance systems to identify individuals.
- **Intrusion Detection**: Helps in detecting unauthorized access or intrusions in restricted areas.

3. Retail

- **Product Categorization**: Automates the process of categorizing products in online stores.
- **Inventory Management**: Monitors stock levels and identifies products in warehouses.

4. Autonomous Vehicles

- **Object Detection**: Identifies and classifies objects such as pedestrians, vehicles, and road signs to navigate safely.
- **Scene Understanding**: Helps in understanding the driving environment for better decision-making.

5. Agriculture

- **Crop Monitoring**: Identifies and classifies crops to monitor their health and detect diseases.
- **Yield Prediction**: Helps in predicting crop yields based on image data.

6. Manufacturing

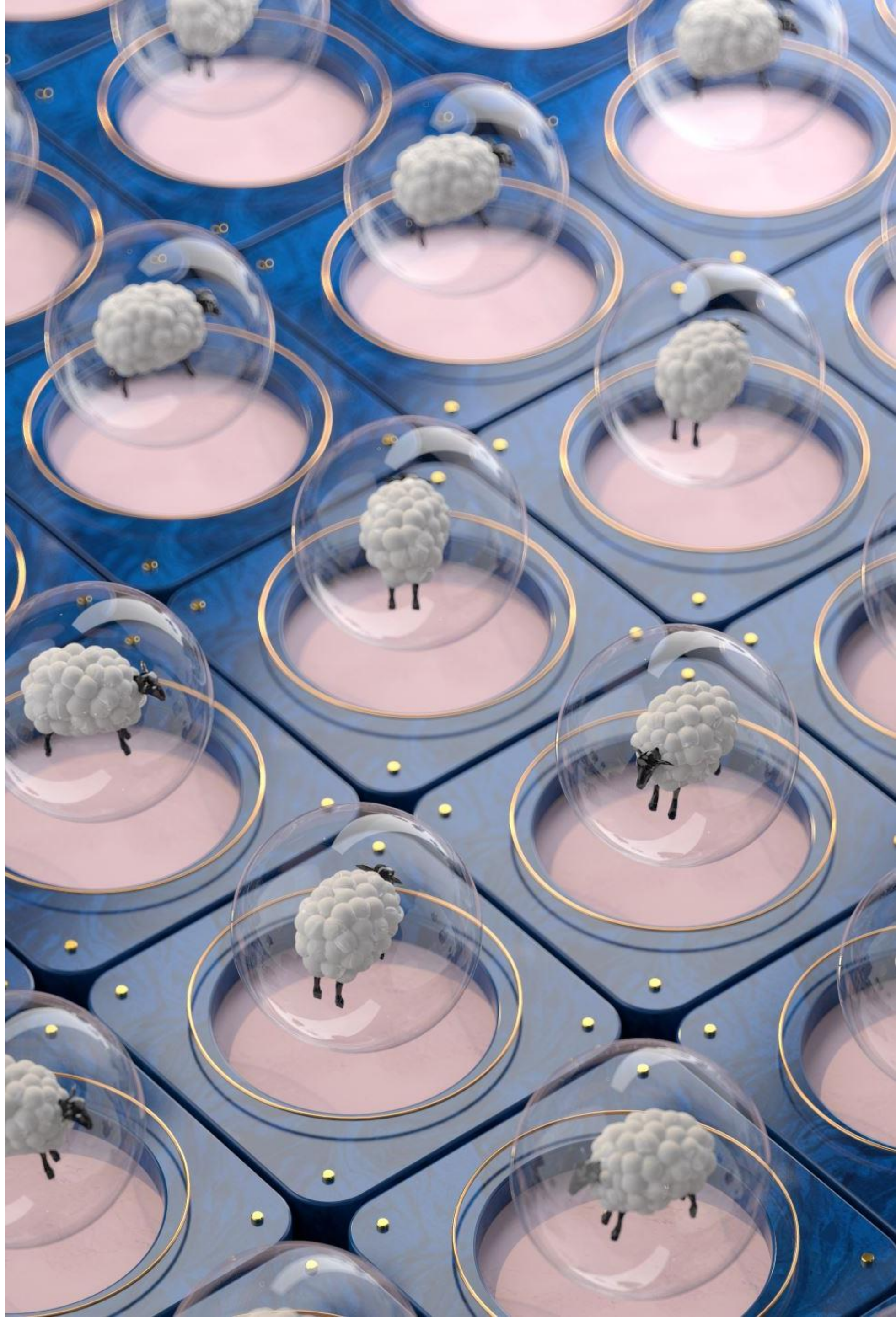
- **Quality Control**: Inspects products for defects and ensures quality standards.
- **Automation**: Automates the classification of parts and products in manufacturing processes.

7. Entertainment

- **Content Moderation**: Classifies and filters inappropriate content in images and videos.
- **Recommendation Systems**: Suggests relevant content based on image classification.

8. Environmental Monitoring

- **Wildlife Conservation**: Identifies and tracks animal species in their natural habitats.
- **Pollution Detection**: Monitors environmental changes and detects pollution levels.



ALGORITHMS AND MODELS OF IMAGE CLASSIFICATION

There isn't one straightforward approach for achieving image classification, thus we will take a look at the two most notable kinds: supervised and unsupervised classification

SUPERVISED CLASSIFICATION

Supervised learning-The algorithm is trained on a labeled image dataset, where the correct outputs are already known and each image is assigned to its corresponding class.

After the training phase, the algorithm uses the knowledge gained from the labeled data to identify patterns and predict the classes of new images.

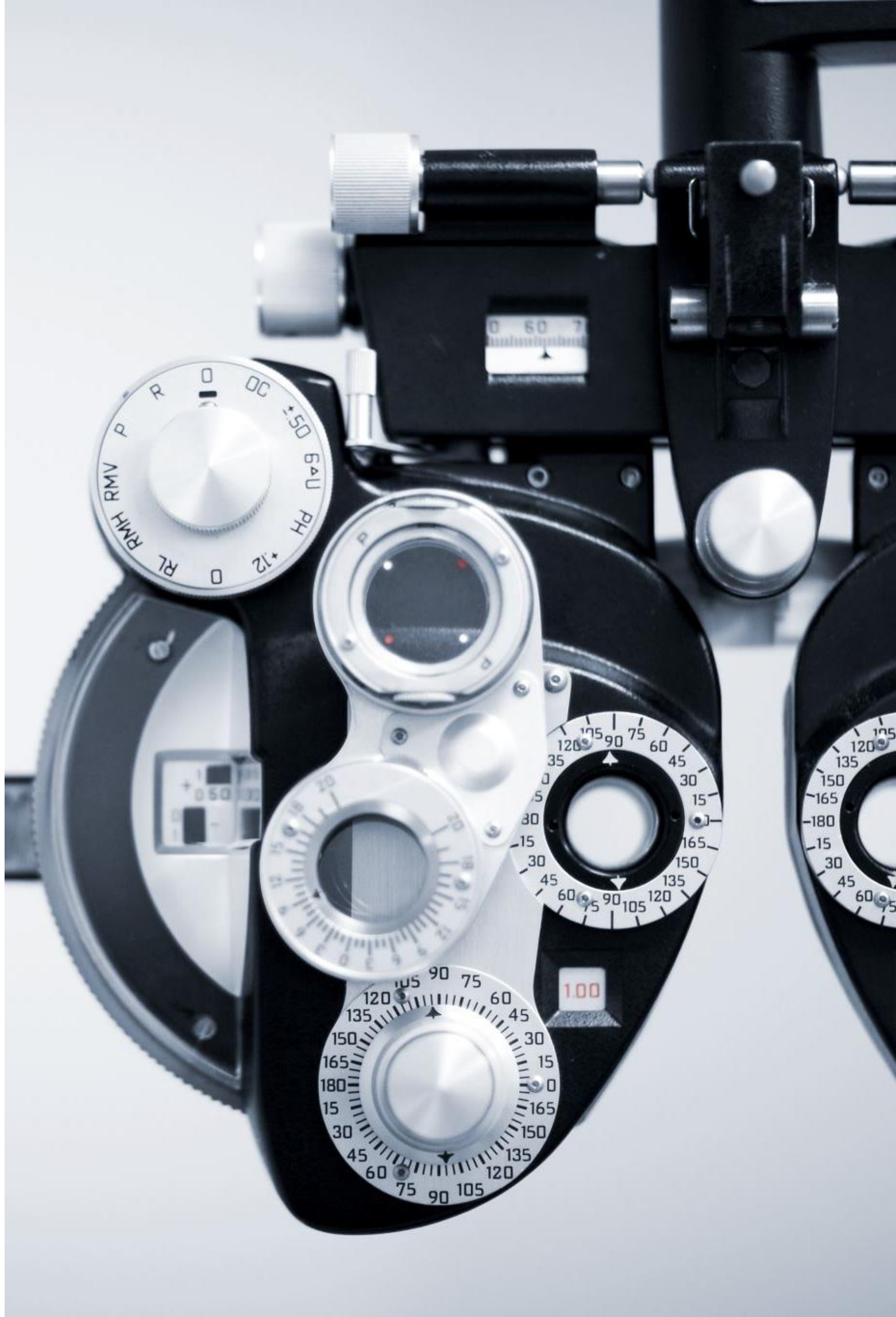
UNSUPERVISED CLASSIFICATION

Unsupervised learning can be seen as an independent mechanism in machine learning; it doesn't rely on labeled data but rather discovers patterns and insights on its own.

The algorithm is free to explore and learn without any preconceived notions, interpreting raw data, recognizing image patterns, and drawing conclusions without human interference.

OBJECT DETECTION

Object detection involves identifying and locating objects within an image. It not only classifies objects but also provides their positions.

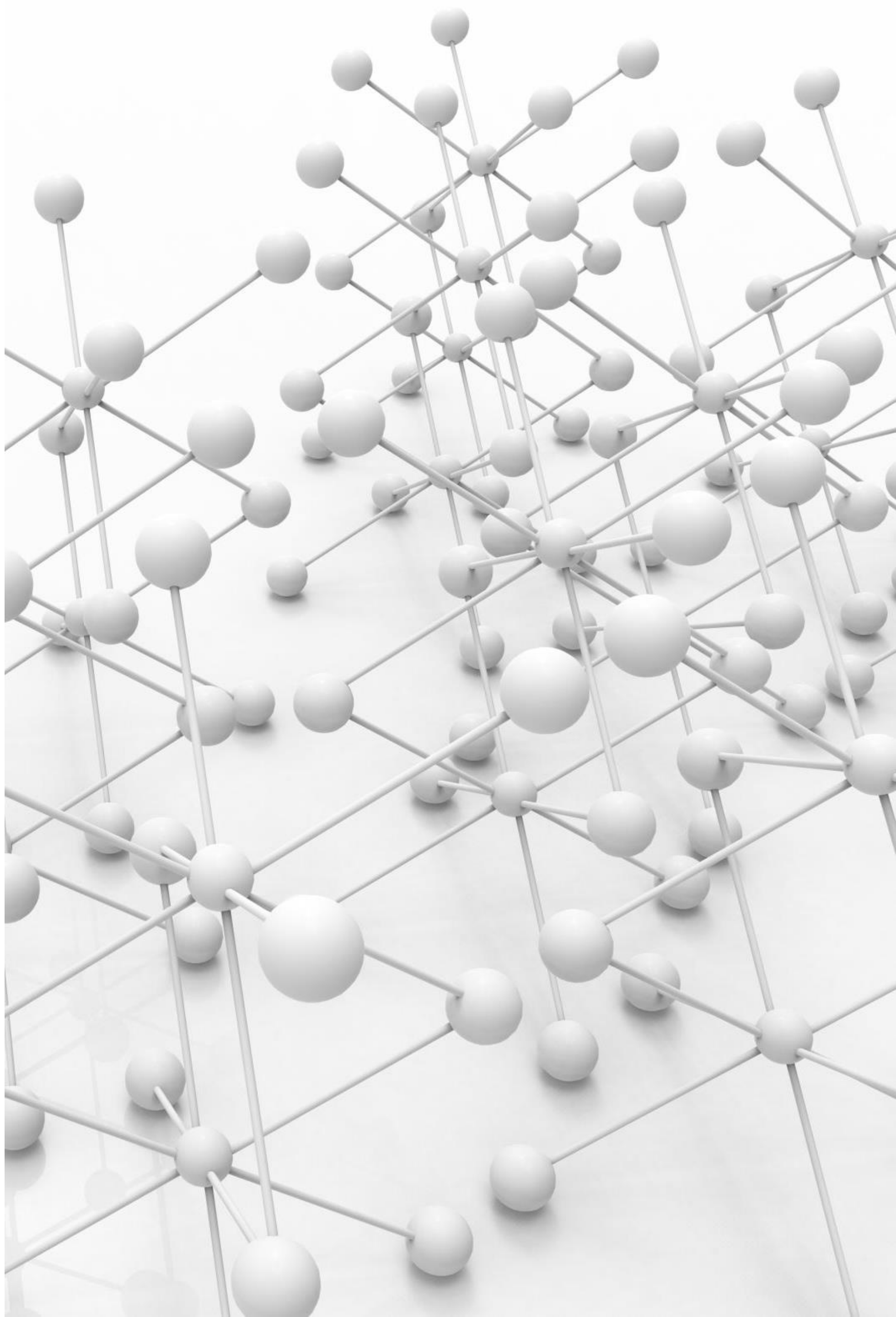


HOW OBJECT DETECTION WORKS

- Feature Extraction: Identifying unique features of objects (e.g., edges, textures).

- Classification: Categorizing objects based on features.

- Localization: Determining the position of objects within an image.



TYPES OF OBJECT DETECTION

SINGLE OBJECT DETECTION

- DETECTING ONE OBJECT IN AN IMAGE.

MULTIPLE OBJECT DETECTION

- DETECTING MULTIPLE OBJECTS SIMULTANEOUSLY.

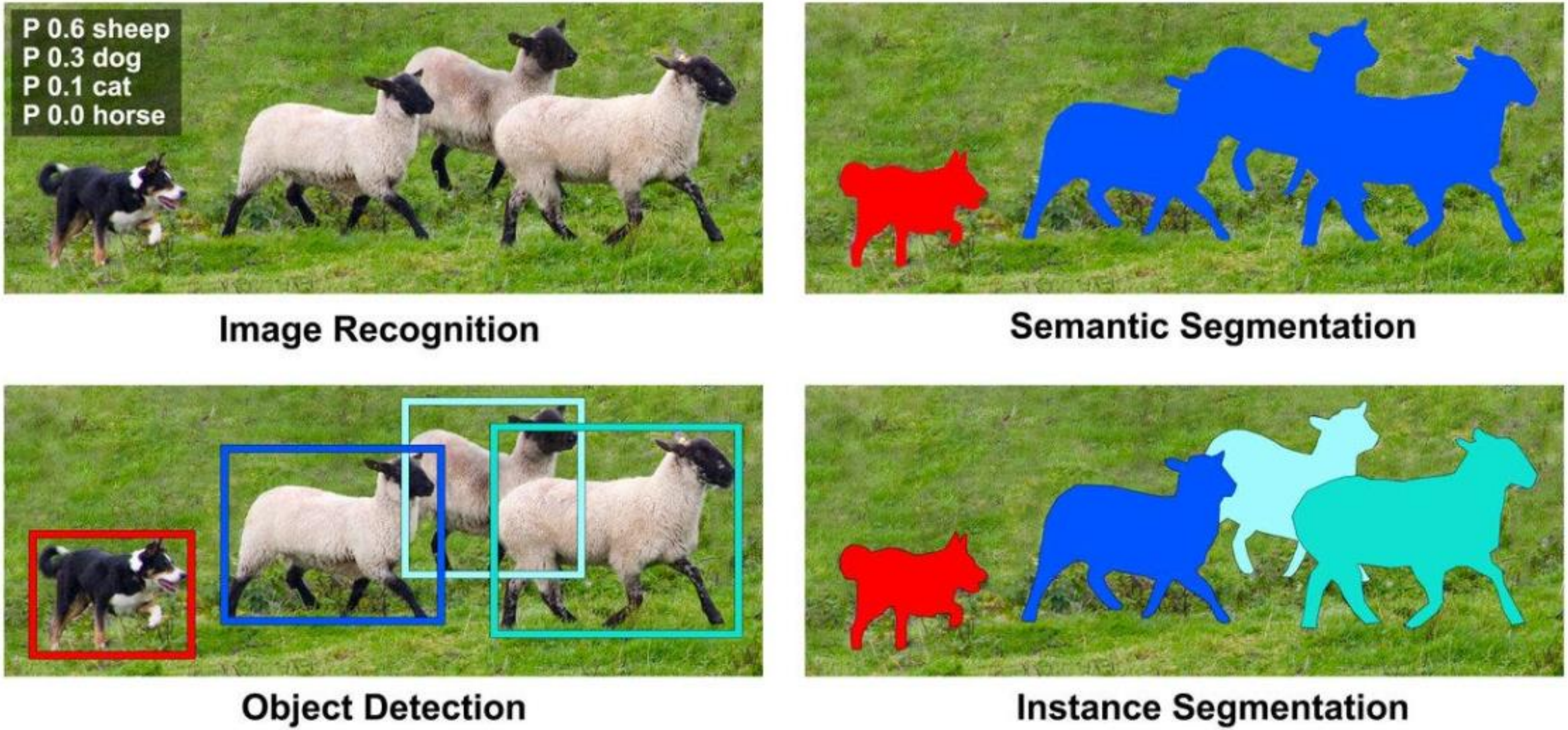
POPULAR MODELS

YOLO (You Only Look Once): Known for its speed and accuracy, YOLO divides images into a grid and predicts bounding boxes and probabilities for each grid cell.

SSD (Single Shot MultiBox Detector): SSD uses a single deep neural network to predict multiple bounding boxes and class scores for objects in the image.

Faster R-CNN: Combines region proposal networks with convolutional neural networks to detect objects more accurately.

- **How Object Detection Works**: It typically involves feature extraction, region proposal, and classification. Models use convolutional neural networks (CNNs) to extract features and predict bounding boxes.
- **Use Cases and Examples**: Object detection is used in self-driving cars to detect pedestrians and other vehicles, in security systems to identify intruders, and in retail to monitor inventory.


P 0.6 sheep
P 0.3 dog
P 0.1 cat
P 0.0 horse
Image Recognition
Semantic Segmentation
Object Detection
Instance Segmentation

Medium.com

3. SEMANTIC SEGMENTATION

Semantic segmentation involves classifying each pixel in an image into a predefined category. Unlike object detection, it provides a more detailed understanding of the image.

- **How Semantic Segmentation Works**: It involves assigning a class label to each pixel in the image. Models use CNNs to learn spatial hierarchies and segment images at the pixel level.
- **Use Cases and Examples**: Semantic segmentation is used in medical imaging to segment organs and tissues, in autonomous driving to understand road scenes, and in agriculture to monitor crop health.

PIXEL-WISE CLASSIFICATION

- **Definition:** In semantic segmentation, pixel-wise classification refers to the process of assigning a specific category or class label to each individual pixel in an image.
- **Example:** Imagine an image of a street scene. Each pixel in the image is classified as belonging to a particular class, such as "road," "car," "pedestrian," "building," etc. This means that every pixel is analyzed and labeled based on what it represents in the real world.

SEGMENTATION MAPS

- **Definition:** A segmentation map is a visual representation of the pixel-wise classification results. It shows how each pixel in the original image has been classified.
- **Example:** Continuing with the street scene example, a segmentation map would display different colors for different classes. For instance, all pixels classified as "road" might be colored blue, "cars" might be colored red, "pedestrians" green, and so on. This map helps to easily visualize which parts of the image correspond to which classes.

IMPLEMENTATION AND TOOLS

Popular frameworks include TensorFlow and PyTorch, which provide pre-trained models and tools for building custom models.

STEPS TO IMPLEMENT OBJECT DETECTION AND SEMANTIC SEGMENTATION

1. **Data Preparation**: Collect and annotate datasets.
2. **Model Selection**: Choose a pre-trained model or build a custom model.
3. **Training**: Train the model on the dataset.
4. **Evaluation**: Evaluate the model's performance using metrics like accuracy.
5. **Deployment**: Deploy the model for inference.

CASE STUDIES

https://github.com/TatevKaren/convolutional-neural-network-image_recognition_case_study

https://medium.com/analytics-vidhya/an-industrial-case-study-on-deep-learning-image-classification-c909d331fc31

RESOURCES

https://medium.com/analytics-vidhya/

https://www.geeksforgeeks.org/what-is-image-classification/